Summary

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Some notes on notation for finding limits:

(a) **The first step in evaluating any limit is to find the form.** This is true, even if the question does not specifically ask for the form.

(b) To find the form of $\lim x \to a \dfrac{f(x)}{g(x)}$, take the limits of the numerator and denominator SEPARATELY. Do not "plug in $x = a$". "Plugging in" can not occur until **after** the function is indicated to be continuous.

(c) When writing a form of a limit, we NEVER write $\lim x \to a \dfrac{f(x)}{g(x)} = \dfrac{0}{0}$. It is the $=$ that makes this a mathematically incorrect statement. Instead, write $\lim x \to a \dfrac{f(x)}{g(x)}$ is of the form $\dfrac{0}{0}$.

(d) When evaluating limits, do NOT drop the $\lim x \to a$ until you have evaluated the limit. For example, the following is INCORRECT:

$$\lim x \to 2 \frac{2x^2 - 4x}{x-2} = \frac{2x(x-2)}{x-2} = 2x = 4$$

Instead, we write:

$$\lim x \to 2 \frac{2x^2 - 4x}{x-2} = \lim x \to 2 \frac{2x(x-2)}{x-2} = \lim x \to 2 2x = 4$$